DHBW Ravensburg

# Intelligence Engineering

## Solving Real World Problems At Scale

Prof. Dr.-Ing. Mark Schutera | October 19, 2026 · *drifting stream*

LECTURE _

# Orientation

what this course is, what it assumes, what yor are going to know after.

You have been given a lot of raw **data**,

# and you want to engineer the shit out of it.

You have been given a business problem,

# and you want to choose the right metrics and objectives to solve it.

Your initial models perform well in offline experiments,

# and you want to deploy them at scale.

You want to harden your deployments

# to quickly detect, debug, and address issues that arise in production.

There is a lot of manual work and handling involved so far,

# and you want to operation vacation your machine learning product or service.

You want to learn about a framework and foundation for building machine learning systems

# that can be shared and reused across machine learning use cases and data products.

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# Overview

## Machine Learning and Data Engineering

# The lecture in one slide

| Lecture | Contents |
|---|---|
| I | Overview of Machine Learning Systems |
| II | Introduction to Machine Learning Systems Design |
| III | Data Engineering Fundamentals |
| IV | Training Data |
| V | Feature Engineering |
| VI | Model Development and Offline Evaluation |
| VII | Model Deployment and Prediction Service |
| VIII | Data Distribution Shifts and Monitoring |
| IX | Continual Learning and Test in Production |
| X | Infrastructure and Tooling for MLOps |
| XI | The Human Side of Machine Learning |

# The lecture in one slide

**I Overview of Machine Learning Systems**

The small part and the big picture

Understanding Machine Learning Systems

**II Introduction to Machine Learning Systems Design**

Business and ML Objectives

Requirements for ML Systems

Iterative Process

Framing ML Problems

Mind Versus Data

**III Data Engineering Fundamentals**

Data Sources

Data Formats

Data Models

Data Storage Engines and Processing

Modes of Dataflow

Batch Processing Versus Stream Processing

# The lecture in one slide

## IV Training Data

Sampling
Labeling
Class Imbalance
Data Augmentation

## V Feature Engineering

Learned Features Versus Engineered Features
Common Feature Engineering Operations
Data Leakage
Engineering Good Features

## VI Model Development and Offline Evaluation

Model Development and Training
Model Offline Evaluation

# The lecture in one slide

**VII Model Deployment and Prediction Service**

Machine Learning Deployment Myths

Batch Prediction Versus Online Prediction

Model Compression

ML on the Cloud and on the Edge

**VIII Data Distribution Shifts and Monitoring**

Causes of ML System Failures

Data Distribution Shifts

Monitoring and Observability

**IX Continual Learning and Test in Production**

Continual Learning

Test in Production

# The lecture in one slide

**X Infrastructure and Tooling for MLOps**

Storage and Compute

Development Environment

Resource Management

ML Platform

Build Versus Buy

**XI The Human Side of Machine Learning**

User Experience

Team Structure

Responsible AI

# Course Work

**Presentation (15 min) + Q&A (open)**

You get recordings of a bee hive entrance, and you are to contribute by data engineering, an annotation policy, a prediction service or specialized tooling.

Open whiteboard, three slides or one page only. Be ready when the class starts.



The bee

# Course Work

What I am looking for ..

**I**ntroduction by Motivation and Literature

**M**ethod by Explanation and or Demonstration

**R**esults show what was achieved and how well

**D**iscussion on findings, limitations and what's next

**Smart and Aesthetic** Thoughts, Designs, Ideas and Questions.




The bee

# Extra Mile

**Special projects**

If you are in for an extra mile, let me know. I have special projects that I can assign to you.

Wrapped in the same presentation and Q&A format, yes it will be about Bananas, and yes it will be more challenging, more time and effort.

Classic loose loose short term, yet **win long term**.



A large banana

The Hive is Real. Dataset on Hugging Face.



It is your obligation to

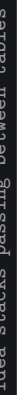
# get yourself a presentation slot and brainstorm a project idea to work on.

# Brainstorm together

**Brainwriting in the big group**

1. Each player writes four ideas.
2. 5 minutes research: look it up online or ask a colleague.
3. Stacks pass to the next table, each player adds two.
4. 5 minutes research.
5. Stacks pass again, each player adds the last two.

**Eight ideas** per player, every stack read at three tables.



LECTURE I

# Overview of a Machine Learning System

# This lecture in one slide

**I Overview of Machine Learning Systems**

The small part and the big picture

Understanding Machine Learning Systems

A **model** is only a small part from the system around it, and some might argue there are a lot of moments when not to reach for machine learning at all.

1

# The small part and the big picture

## Don’t confuse the model with the machine learning system.

# Enabling the delivery of the logic

The **model** is the small part of a machine learning system. With foundation models and AI engineering, that part got smaller still. The essential work of a machine learning system remains and stacks around business understanding, infrastructure, design pattern and data engineering.

Before learning about solving a problem with machine learning, let's first assess when it is actually a good idea to do so.

# (1) Learn

The problem you are solving, requires your system to have the capability to **learn**. Learning can look different depending on the problem and its context and boundary conditions. You already know about supervised learning, unsupervised learning and reinforcement learning paradigms.

Machine learning is an approach to (1) learn (2) complex patterns from (3) existing data and use these patterns to make (4) predictions on (5) unseen data.

# (2) Pattern

Learning requires the existence of a **pattern**, as opposed to noise or randomness. It is not trivial nor obvious whether there is a pattern in the data, and whether it is learnable. Complex pattern is more difficult to embed or state explicitly, this is where machine learning has an edge over classical approaches.

Machine learning is an approach to (1) learn (2) complex patterns from (3) existing data and use these patterns to make (4) predictions on (5) unseen data.

# (3) Data

Data is available, accessible or acquirable. The data is of **sufficient** quality and quantity to support learning, (and to validate for that matter). If your approach does not require data, it is because you build on embeddings or foundation models, which themselves are built on data in the first place, or you need to rely on data acquisition and curation as part of your system.

Machine learning is an approach to (1) learn (2) complex patterns from (3) existing data and use these patterns to make (4) predictions on (5) unseen data.

# (4) Predictions

Machine learning is about making predictions, **always**. Predictions can be of different types, such as classification, regression, ranking, clustering, generation and on and on. By reframing problems in terms of predictions, we can leverage machine learning to solve them. Predictions are approximations by nature and therefore come with uncertainty, which needs to be accounted for.

Machine learning is an approach to (1) learn (2) complex patterns from (3) existing data and use these patterns to make (4) predictions on (5) unseen data.

# (5) Unseen Data

In a closed system or domain, the data is finite and the system can be fully specified by optimization. However, often the domain is open and infinite, and the system needs to be able to **generalize** to unseen data. Of course, the unseen data is not completely unknown, it is drawn from the same distribution (i.i.d.) as the training data and is thus able to generalize to the unseen data inside a specific domain or distribution.

Machine learning is an approach to (1) learn (2) complex patterns from (3) existing data and use these patterns to make (4) predictions on (5) unseen data.

# Catalysts

A catalyst is a substance that increases the rate of reaction without undergoing change itself.

**The task is repetitive.**

In a sense that we have a lot of data to learn a pattern from.

**It scales.**

The need to scale a specific prediction capability, not only a few times.

**The context changes.**

Meaning that hard coded or other solutions will be brittle and break.

**Being wrong is not too bad.**

Approximations on unseen data. Failures need to be manageable.

# When Not to Use Machine Learning

**It is unethical.**

Well, that might also be a general rule. But doing stuff at scale, without human oversight, has its implications.*

**There's a simple solution.**

A simple solution, heuristic, or baseline should always be the first step regardless.

**It's not economically viable.**

The value of the prediction, needs to be greater than the cost of the system, including development, deployment and maintenance.

* More? Enroll to my Philosophy of AI course.

Describe two use-cases and the rationale for whether or not to use machine learning.

# One Machine Learning Use-Case and One which is not.

# The field is moving, 2026

Adoption is near universal, value capture is not.

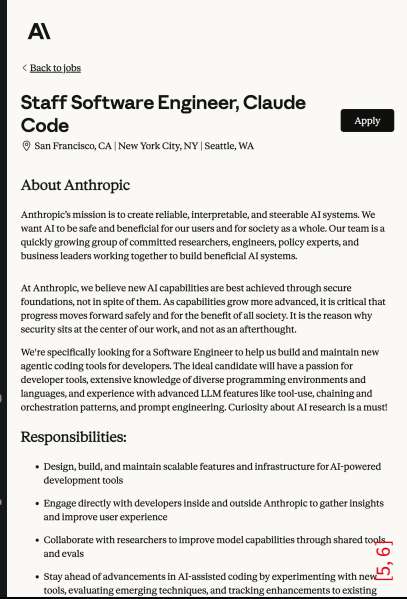
- **Adoption** 88% of organizations use AI in at least one function, 72% use gen AI.
- **Reach** Gen AI reached 53% of the population in three years, faster than the PC or the internet.
- **Productivity** 80% of users report personal gains.
- **Slop** 40% of desk workers say they received AI workslop in the past month. Each case costs almost two hours to fix, about $186 per employee per month.
- **Value** Only **37%** of organizations attribute any EBIT impact to AI, flat since 2025, and 6% qualify as high performers.

[2–4]

# “… in twelve months, we may be in a world where AI is writing essentially all of the code.”

**Dario Amodei, March 2025. Nineteen months later, Anthropic lists 94 open software engineer roles.**

Anthropic hiring, October 2026



‹ Back to jobs

**Staff Software Engineer, Claude Code**

Apply

San Francisco, CA | New York City, NY | Seattle, WA

About Anthropic

Anthropic's mission is to create reliable, interpretable, and steerable AI systems. We want AI to be safe and beneficial for our users and for society as a whole. Our team is a quickly growing group of committed researchers, engineers, policy experts, and business leaders working together to build beneficial AI systems.

At Anthropic, we believe new AI capabilities are best achieved through secure foundations, not in spite of them. As capabilities grow more advanced, it is critical that progress moves forward safely and for the benefit of all society. It is the reason why security sits at the center of our work, and not as an afterthought.

We're specifically looking for a Software Engineer to help us build and maintain new agentic coding tools for developers. The ideal candidate will have a passion for developer tools, extensive knowledge of diverse programming environments and languages, and experience with advanced LLM features like tool-use, chaining and orchestration patterns, and prompt engineering. Curiosity about AI research is a must!

Responsibilities:

- Design, build, and maintain scalable features and infrastructure for AI-powered development tools
- Engage directly with developers inside and outside Anthropic to gather insights and improve user experience
- Collaborate with researchers to improve model capabilities through shared tools and evals
- Stay ahead of advancements in AI-assisted coding by experimenting with new tools, evaluating emerging techniques, and tracking enhancements to existing

[5, 6]

2

# Understanding Machine Learning Systems

## Research and Production are not the same.

# Horizontal versus Vertical

Nothing motivates system thinking more than the need to go horizontal. Academia is vertical, with a narrow focus on a single problem, where drilling deep, fast is rewarded and focus allows to teach a wide range of concepts.

| | Research | Production |
|---|---|---|
| **Objectives** | State of the art, highly engineered | Stakeholder outcome driven |
| **Compute** | Training and high throughput | Fast inference with low latency |
| **Data** | Often static benchmark oriented | Shifting domains and structures |
| **Usability metrics** | Usually not considered | UI, Fairness, Robust, Interpretability |

# Stakeholders and Objectives

In research the objective is one number on one benchmark. Ship the same model as a product and five people want five different things from it: **conflicting requirements** are the norm, and not all of them are must haves.

ML engineers | Platform team | Product team | Sales team | Manager



The bee hive

Different stakeholders have different objectives and think in different solution spaces.

# Define the next step for the bee hive and line out how to reach that.

# Five people, five solution spaces

Each of them is right. The **objective** is what they are measured on, the solution space is where they will look for it.

| | Objective | Solution space |
|---|---|---|
| ML engineers | better bee counts | more labelled frames, a bigger model |
| Platform team | nothing pages at 3am | freeze the model, build the platform |
| Product team | the hive view feels live | a latency budget of 100 ms |
| Sales team | revenue per hive | an alert for every hive, sold as premium |
| Manager | margin | fewer models, and maybe fewer engineers |

# Computational priorities

Research optimises **throughput**, production optimises **latency**, and one buys the other.

| Processing | Latency | Throughput |
|---|---|---|
| One query at a time | 10 ms | 100 queries/s |
| Batches of 10 | 20 ms | 500 queries/s |
| Batches of 100 | 50 ms | 2,000 queries/s |

Every row is the same model on the same hardware. Only the batch changed.

Batching waits for the queue to fill, so each single request waits longer while the machine as a whole serves far more of them.

Training leans throughput: nobody is waiting on one gradient step. Serving leans latency: somebody is.

Batch size is not the only lever. Model size, sample size and how many models sit in one request all move both numbers, and they do not move them in the same direction.

[1]

# Latency is a distribution

Ten requests, latency in ms:

10.0, 9.7, 10.4, **90.0**, 9.9, 10.0, 11.0, 9.5, 10.2, 10.0

The mean is 18.1 ms. Nine of the ten requests were served in about 10 ms and not one of them took 18. The average describes a request that never happened.

Users get less patient.

2017, Akamai: a 100 ms delay can cost 7% of conversions.

2019, Booking.com: about 30% more latency cost about 0.5% in conversion, “a relevant cost for our business”.

2016, Google: over half of mobile users leave a page that takes more than three seconds.

[6-7]

# Percentiles, not averages

The same ten requests, sorted. A percentile is a kind of promise: $q$ percent of requests finish in $p_q$ or less.

9.5, 9.7, 9.9, 10.0, 10.0, 10.0, 10.2, 10.4, 11.0, **90.0**

$p_{50}$ $p_{90}$ $p_{99}$

$$p_q = \min\left\{ t : \frac{\#\,\text{requests} \le t}{n} \ge \frac{q}{100} \right\}$$

The mean of 18.1 ms moves with the outlier, the percentile makes the outlier visible.

A rule of thumb: the band between $p_{90}$ and $p_{99}$ shares a cause you can find, above $p_{99}$ it is often one broken machine.

With ten requests $p_{99}$ is just the maximum. It takes 100 requests before $p_{99}$ and the maximum can differ.

Quality gates count in nines: three nines is $p_{99.9}$ (1,000 requests), five nines $p_{99.999}$ (100,000). As uptime, 8.8 hours against 5.3 minutes down a year.

[10]

# Any metric is a distribution

Latency is a distribution. As is any metric. So if you are only reporting accuracy, you are **underdelivering** already.




One model, 10,000 random seeds: CIFAR 10 accuracy moves by 1.82 points. A single run is a single draw.

Start with the questions the system has to answer. Pick the metrics that answer them. Only then build the methods that move those metrics.

[11]

# Data in research

A research dataset is a teaching artefact. Somebody already did the work.

- Clean and well formatted
- Static, so the community can benchmark against it
- Quirks of the dataset are known and documented
- Open source scripts to feed your models
- Historical, already stored somewhere

Titanic [12]: 891 rows, 12 columns, and every one of its quirks is written up in a tutorial somewhere.

Karpathy: at Tesla you lose sleep over data




[13]

# Leaderboards and Benchmarks

- **System** A leaderboard scores a **model**, and a system is not a model.
- **Metric** One metric on one frozen test set. Whether it measures what it claims is rarely argued.
- **Overfit** Every run against a benchmark stales it. GLUE was saturated within a year.
- **Chance** Scanning 10,000 seeds moves CIFAR 10 by 1.82 points, nothing else changed.

GLUE 2018. Trust me bro, nine tasks

**GLUE: A Multi-Task Benchmark and Analysis Platform for Natural Language Understanding**

**Alex Wang**[1], **Amanpreet Singh**[1], **Julian Michael**[2], **Felix Hill**[3], **Omer Levy**[2], **and Samuel R. Bowman**[1]

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Human ability to understand language is *general*, *flexible*, and *robust*. In contrast, most NLU models above the word level are designed for a specific task and struggle with out-of-domain data. If we aspire to develop models with understanding beyond the detection of superficial correspondences between inputs and outputs, then it is critical to develop a unified model that can execute a range of linguistic tasks across different domains.

To facilitate research in this direction, we present the General Language Understanding Evaluation (GLUE, gluebenchmark.com): a benchmark of nine diverse NLU tasks, an auxiliary dataset for probing models for understanding of specific linguistic phenomena, and an online platform for evaluating and comparing models. For some benchmark tasks, training data is plentiful, but for others it is limited or does not match the genre of the test set. GLUE thus favors models that can represent linguistic knowledge in a way that facilitates sample-efficient learning and effective knowledge-transfer across tasks. While none of the datasets in GLUE were created from scratch for the benchmark, four of them feature privately-held test data, which is used to ensure that the benchmark is used fairly.

We evaluate baselines that use ELMo (Peters et al., 2018), a powerful transfer learning technique, as well as state-of-the-art sentence representation models. The best models still achieve fairly low absolute scores. Analysis with our diagnostic dataset yields similarly weak performance over all phenomena tested, with some exceptions.

**The GLUE benchmark** GLUE consists of nine English sentence understanding tasks covering a broad range of domains, data quantities, and difficulties. As the goal of GLUE is to spur develop-

| Corpus | \|Train\| | Task | Domain |
|---|---|---|---|
| Single-Sentence Tasks | | | |
| **CoLA** | 8.5k | acceptability | misc. |
| SST-2 | 67k | sentiment | movie reviews |
| Similarity and Paraphrase Tasks | | | |
| MRPC | 3.7k | paraphrase | news |
| STS-B | 7k | textual sim. | misc. |
| **QQP** | 364k | paraphrase | online QA |
| Inference Tasks | | | |
| **MNLI** | 393k | NLI | misc. |
| QNLI | 108k | QA/NLI | Wikipedia |
| RTE | 2.5k | NLI | misc. |
| **WNLI** | 634 | coref./NLI | fiction books |

Table 1: Task descriptions and statistics. **Bold** denotes tasks for which there is privately-held test data. All tasks are binary classification, except STS-B (regression) and MNLI (three classes).

quire models to share substantial knowledge (e.g., trained parameters) across tasks, while maintaining some task-specific components. Though it is possible to train a model per task and evaluate the resulting set of models on this benchmark, we expect that inclusion of several data-scarce tasks will ultimately render this approach uncompetitive.

The nine tasks include two tasks with single-sentence inputs: Corpus of Linguistic Acceptability (CoLA; Warstadt et al. 2018) and Stanford Sentiment Treebank (SST-2; Socher et al. 2013). Three tasks involve detecting semantic similarity: Microsoft Research Paraphrase Corpus (MRPC; (Dolan and Brockett, 2005)), Quora Question Pairs[1] (QQP), and Semantic Textual Similarity Benchmark (STS-B; Cer et al. 2017). The remaining four tasks are formatted as natural language inference (NLI) tasks, such as the Multi-Genre NLI corpus (MNLI; Williams et al. 2018) and Recog-

# Data in production

Nobody did the work. The data is a by-product of a business that was not run for your benefit.

- Messy, noisy, possibly unstructured
- **Likely biased, and you do not know how**
- Labels sparse, imbalanced, or incorrect
- Changing requirements can mean relabeling
- Privacy and regulatory concerns
- Generated by users, systems, third parties

# LLM as a judge

An LLM judge grades your model with another model. Against its labels $\tilde{y}$ instead of the true $y$, your accuracy is off by as much as the judge is wrong.

Never for **validation**. For training data at most, and only while it adds no more $\tilde{y}$ than your ground truth already has.

- **Hallucination** grades claims it cannot check
- **Position bias** swap the answers, 65% of verdicts hold
- **Self-preference** rates its own, and its family's, outputs higher




The judge meets the judged [17–19]

# Perpetuating Bias

Machine learning algorithms do not predict the future, they encode the past, and then they act on it.

**The invisible hand** of performative prediction at scale: the prediction is not a reading of the world, it is a push on it.

2014 to 2018, Amazon built a résumé ranker on ten years of its own hiring. It learned to penalise the word “women’s” and graduates of two women’s colleges. The project was scrapped.

2015, Google Photos labelled two Black users as gorillas. The fix shipped, and three years later, was still to delete the label from the vocabulary.

[20–23]

# Interpretability

Research rewards performance, not interpretability. Industry asks for it anyway, from two directions:

- **Users**, business leaders and end users alike, to see why a decision was made, so they can trust it and catch what is wrong with it.
- **Developers**, to debug and improve the model at all.

Most of us use a fridge without understanding it. Fewer feel that way about the thing deciding on their mortgage. Only 19% of large companies are working on explainability.

Jev, 2026, is built only to decide: a typed answer with a probability, and no text to read an explanation from.

Press 2: hand Claude recursive permission




[24]

# Human on the loop

Putting a human on the loop is the standard answer to an uninterpretable model. The evidence that it works is **thin**.

Supervising reliable automation degrades exactly the capability being supervised: vigilance falls, response to the first failure is poor, and the takeover you designed for arrives too late.

Tempe, 2018: the system saw the pedestrian 5.6 s before impact and handed over at 1.2 s. The safety driver, eyes on her phone for 34% of the drive, looked up 1 s before. Test a human on the loop like any other design decision.



Tempe Police dashcam, March 2018

# Why this is different from traditional software engineering

Traditional software engineering tests and versions **code**. A machine learning system has to test and version code, **data**, and everything the two produce together.

- How do you version a dataset, and diff two versions of one?
- How do you deploy into an environment that keeps moving?
- How do you monitor and debug a system with no stack trace?
- How do you avoid being harmful at scale?

Two books worth owning,

**Software Engineering at Google** [26]: engineering is programming integrated over time. Every question on the left is a question about time.

**The Mythical Man-Month** [27]: essential complexity against accidental. Data is essential complexity, and there is no silver bullet for it either.

**A trained model is the small box in the middle.**

You might argue that the engineering and development side of machine learning is more exciting.

# Most jobs will be, and already are in productionizing machine learning systems.

This is the last slide.
Look how far you've come.

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